

Zigzags and Triangles

Fall 2021 ARML Power Contest

The Background

There are a number of ways in which triangular arrays of numbers lead to interesting patterns. Successful mathletes are certainly aware of Pascal's triangle (also known as Tartaglia's triangle, Yanghui's triangle, and several others who studied it long before Pascal!). This contest examines some other ways to combine numbers in a triangular fashion.

Consider a sequence of real numbers $\{a_n\}$, which may be finite or infinite. Its *corresponding triangular array* is created from the sequence as follows:

Step 1: Place the sequence along the left-hand edge of a triangular array.

Step 2: To compute any other entry in the array add the entry to its left and to its above-left.

Step 3: Complete as many rows of the array as desired.

The array will start as follows:

$$\begin{array}{ccccccc}
 & & & & a_0 & & \\
 & & & & \searrow & & \\
 & a_1 & & \longrightarrow & a_0 + a_1 & & \\
 & & \searrow & & \searrow & & \\
 a_2 & \longrightarrow & a_1 + a_2 & \longrightarrow & a_0 + 2a_1 + a_2 & &
 \end{array}$$

For a concrete example, for the sequence $a_0 = 0, a_1 = 1, a_2 = 2, \dots$, the corresponding triangular array's first few rows will be

$$\begin{array}{cccccccc}
 & & & & 0 & & & \\
 & & & & 1 & & 1 & \\
 & & 2 & & 3 & & 4 & \\
 & 3 & & 5 & & 8 & & 12 \\
 4 & & 7 & & 12 & & 20 & & 32 \\
 5 & 4 & 9 & 16 & 28 & 48 & 80 &
 \end{array}$$

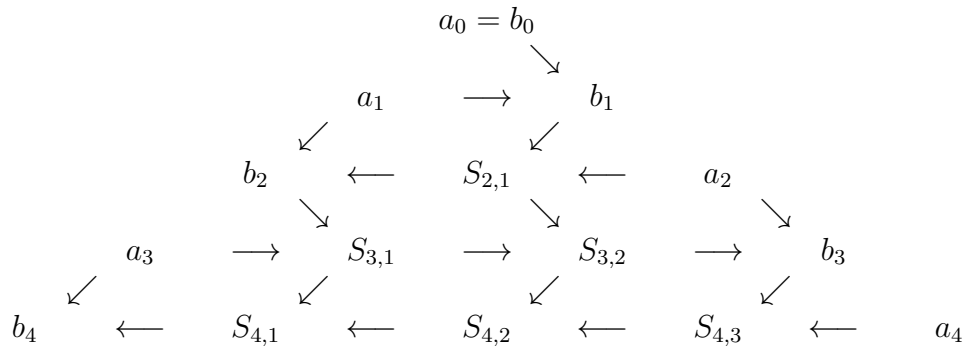
The entries in such a triangular array are numbered $T_{n,k}$ where n is the row number, starting with zero, and k is the position from the left, also starting at zero. So, for example, in the triangular array above $T_{4,3} = 20$. In the corresponding triangular array created from the sequence $\{a_n\}$, the entries $T_{n,0} = a_n$ by definition, since these numbers form the left-hand edge of the triangular array. In fact, the triangle can be thought of as being created by the recursive definition

$$T_{n,k} = \begin{cases} a_n & k = 0 \\ T_{n,k-1} + T_{n-1,k-1} & 1 \leq k \leq n \end{cases}$$

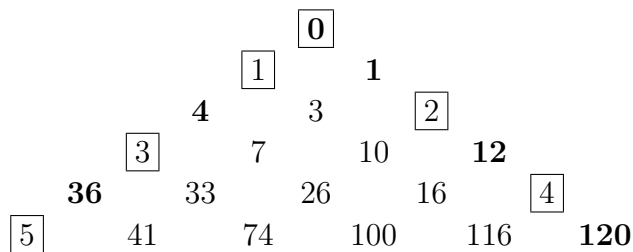
Given a sequence $\{a_n\}$ its *triangular transform* is the sequence $\{b_n\}$ defined by $b_n = T_{n,n}$. That is, b_n is the entry at the right-hand edge of row n of the triangular array corresponding

to the sequence $\{a_n\}$. Thus, the triangular transform of the sequence $0, 1, 2, 3, 4, 5, \dots$ is $0, 1, 4, 12, 32, 80, \dots$.

A different way of using a triangular array to transform a sequence leads to some fascinating results. Instead of placing the sequence $\{a_n\}$ on the left-hand edge of the triangular array, alternately place terms on the left- and right-hand edges. In rows where a_n has been placed on the left-hand edge, the entries to its right are computed in the above manner, adding the entry to the left and to the above left. In rows where a_n has been placed on the right, the process is backward, adding the entries to the right and above right. Schematically this is shown below:



The transformed sequence consists of the terms at the opposite ends of the rows from where the $\{a_n\}$ terms were placed, as indicated by the b 's in the above triangular array. This triangular array is called the *boustrophedon array* and the resulting transformed sequence is called the *boustrophedon transform* of the original sequence. The word “boustrophedon” comes from the Greek roots *bous*, meaning “ox,” and *strophos*, meaning “turning,” and refers to the zigzag path that an ox takes while walking back and forth, plowing a field. The boustrophedon transform of the sequence $0, 1, 2, 3, 4, 5, \dots$ is $0, 1, 4, 12, 36, 120, \dots$, as shown below, where the original sequence is boxed and the transformed sequence is in bold:



A *permutation* is an arrangement of the first n positive integers. For example, a permutation of the first five integers could be $\langle\langle 5, 3, 1, 4, 2 \rangle\rangle$ or even $\langle\langle 1, 2, 3, 4, 5 \rangle\rangle$. The $\langle\langle \rangle\rangle$ notation is special to this contest only, to distinguish between permutations and arbitrary sequences.

A permutation is called an *up-down* permutation if the numbers alternately increase and decrease, with the first step being increasing. A *down-up* permutation has the numbers alternately increasing and decreasing with the first step being decreasing. So $\langle\langle 2, 4, 1, 3 \rangle\rangle$ is an up-down permutation while its reverse, $\langle\langle 3, 1, 4, 2 \rangle\rangle$ is a down-up permutation. An *alternating* permutation is one that is either up-down or down-up.

The Problems

There are 40 points available on this contest. The number of points each problem or part of a problem is worth is listed at the end of the problem as well as on the scoresheet.

In the problems that follow, the notation $\{c_n\}$ will be used as shorthand to denote a sequence $c_0, c_1, c_2, \dots$. It will *never* denote a set in this contest. The sequences in this contest will *always* start at index 0. They may be finite or infinite, and if finite can be as short as one term (i.e., just the single term c_0). The notation $\langle\langle\rangle\rangle$ will be specifically used to indicate permutations of the first n positive integers, such as $\langle\langle 5, 3, 1, 4, 2 \rangle\rangle$.

The problems concern both types of arrays and transforms discussed in the background material. Please be sure to respond to each problem with the correct type of array or transform in mind!

1. For each sequence below, list rows 0 through 5 of the corresponding triangular array, and list terms 0 through 5 of its triangular transform.
 - (a) $1, 1, 1, 1, 1, 1, \dots$ [2 pts.]
 - (b) $1, -1, 1, -1, 1, -1, \dots$ [2 pts.]
2. List terms 0 through 5 of a sequence whose triangular transform is $0, 1, 2, 3, 4, 5, \dots$ [2 pts.]
3. For each sequence below, list rows 0 through 5 of the boustrophedon array, and list terms 0 through 5 of its boustrophedon transform. (Do not box or underline entries—that highlighting was for demonstration purposes only.)
 - (a) $1, 1, 1, 1, 1, 1, \dots$ [2 pts.]
 - (b) $1, -1, 1, -1, 1, -1, \dots$ [2 pts.]
4. List terms 0 through 5 of a sequence whose boustrophedon transform is $0, 1, 2, 3, 4, 5, \dots$ [2 pts.]
5. Given a sequence $\{a_n\}$, determine a recursive definition for the entries of its boustrophedon array. [2 pts.]
6. Given the sequence $\{a_n\}$, prove that each entry of the corresponding triangular array is given by the formula $T_{n,k} = \sum_{i=0}^k \binom{k}{i} a_{n-i}$ where $\binom{k}{i}$ is the usual binomial coefficient $\binom{k}{i} = \frac{k!}{i!(k-i)!}$. You may use standard facts about the binomial coefficients in your proof. [3 pts.]
7. Given a sequence $\{b_n\}$, prove that there is a unique sequence $\{a_n\}$ whose triangular transform is $\{b_n\}$. The sequence $\{a_n\}$ will be called the *inverse triangular transform* of $\{b_n\}$. [2 pts.]

8. Given a sequence $\{b_n\}$ of integers, prove that its inverse triangular transform is also a sequence of integers. [1 pt.]
9. Given a sequence $\{b_n\}$, prove that there is a unique sequence $\{a_n\}$ whose boustrophedon transform is $\{b_n\}$. The sequence $\{a_n\}$ will be called the *inverse boustrophedon transform* of $\{b_n\}$. [2 pts.]
10. Prove that the inverse boustrophedon transform of a sequence of integers is again a sequence of integers. [1 pt.]

Alternating permutations and the boustrophedon transform

11. There are five up-down permutations of the first 4 positive integers. List them. [2 pts.]
12. Compute the number of up-down permutation of the first 5 positive integers. [2 pts.]
13. Prove that the number of up-down permutations of the first n positive integers and the number of down-up permutations of the first n positive integers are equal. [2 pts.]
14. Let A_n be the number of up-down permutations of the first n positive integers. For example, there are two up-down permutations of the first three positive integers: $\langle\langle 1, 3, 2 \rangle\rangle$ and $\langle\langle 2, 3, 1 \rangle\rangle$. The sequence $\{A_n\}$ begins $1, 1, 1, 2, 5, \dots$ where the next term is the correct answer to the previous problem, and the “empty permutation” of the first 0 positive integers is declared by definition to be an up-down permutation. Prove that A_n satisfies the recurrence relation

$$2A_{n+1} = \sum_{k=0}^n \binom{n}{k} A_k A_{n-k}$$

(with, of course, the initial value $A_0 = 1$). [3 pts.]

15. Use the above recurrence to compute A_6 . [1 pt.]

The boustrophedon triangle for the sequence $1, 0, 0, 0, \dots$ is called the *Seidel-Entringer-Arnold triangle* (after three mathematicians who studied it independently). The *Entringer numbers* $E_{n,k}$ are the entries of this triangle, where the even numbered rows are read *backward*. That is, if the entries of the boustrophedon array for $1, 0, 0, 0, \dots$ are denoted $S_{n,k}$ then

$$E_{n,k} = \begin{cases} S_{n,k} & n \text{ is odd} \\ S_{n,n-k} & n \text{ is even} \end{cases}.$$

16. Show that the Entringer numbers satisfy the recursive definition

$$E_{n,k} = \begin{cases} 1 & n = k = 0 \\ 0 & n > 0, k = 0 \\ E_{n,k-1} + E_{n-1,n-k} & k > 0 \end{cases}.$$

[2 pts.]

17. Prove that $E_{n,k}$ is the number of down-up permutations of the first $n + 1$ positive integers whose first number is exactly $k + 1$. For example, $E_{3,2} = 2$ and there are exactly 2 down-up permutations of the first $3 + 1 = 4$ integers whose first number is $2 + 1 = 3$, namely $\langle\langle 3, 1, 4, 2 \rangle\rangle$ and $\langle\langle 3, 2, 4, 1 \rangle\rangle$ [3 pts.]
18. Prove that $A_n = E_{n,n}$. [2 pts.]

There are no problems past this point. The following is additional history and mathematical connections.

Alternating permutations were first studied by Désiré André around 1880. This contest names the sequence $\{A_n\}$ in his honor. The problem of determining the number of alternating permutations (which is, of course, $2A_n$ by the results of these problems) is called *André's problem*.

The numbers A_n are called Euler numbers, zigzag numbers, or up/down numbers. Be careful, because there are other sequences of numbers also called Euler numbers. The even-indexed A 's are also known as secant numbers, while the odd-indexed A 's are called tangent numbers. This is because they are related to the Maclaurin series for the secant and tangent functions. (Maclaurin series are studied in calculus.) These numbers are related to several other kinds of important numbers, such as Bernoulli and Stirling numbers.

A neat result about these numbers that can be proven from their relationship to the secant and tangent functions is that when n is large, the ratio A_n/A_{n+1} is very close to $n\pi/2$ —note that $\pi/2$ is the location of an asymptote of the secant and tangent functions!

There are formulas for computing the terms of the boustrophedon transform or the inverse boustrophedon transform of a sequence directly in terms of the sequence, the up/down numbers, and the binomial coefficients.

More detailed information about these connections can easily be found by searching the web.